



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : PHY F213
Course Title : OPTICS
Instructor-in-Charge : Niladri Sarkar

1. Course Description: The course will cover Geometrical optics, wave optics, Diffraction, Laser theory, and the recent trends of research in the optical physics.

2. Scope and Objective of the Course:

The objective of the course is to give an introduction to basic phenomena in optics and the techniques used to deal with them. The recent development of the subject and its application in the research level will be discussed. Course will assume a basic knowledge of optics at the level of the core Physics courses. The course will also provide theoretical background for the optics experiments done in Electromagnetism & Optics Lab. (PHYF214).

3. Text Books:

Optics: Ajoy Ghatak, 8th Edition, Tata McGraw Hill

4. Reference Books:

1. Fundamentals of Optics by F.A. Jenkins and H.E. White, McGraw Hill Book Co., 3rd Ed.
2. Optical Physics by A. Lipson, S. G. Lipson, H. Lipson Cambridge University Press, 4th Ed.
3. Optics by Eugene Hecht, A. R. Ganesan, Pearson, 4th Ed.
4. Introduction to Optics by F. L. Pedrotti, L. S. Pedrotti, L. M. Pedrotti, Pearson, 3rd Ed.

5. Course Plan:

Module No.	Learning Objectives	Topics to be covered	Reference Chapters and Sections	Learning outcome
1	Introduction to optics, Fermat's principle.	Basic introduction to optics, Newton's corpuscular model, laws of reflection and refraction from Fermat's Principle and its applications	T.B.: 2.1-2.5	To derive laws of reflection & refraction using Fermat's principle. Modern version of Fermat's Principle and spherical refracting surfaces. Understanding various optical phenomena for example mirage by solving the ray equation, obtain ray paths in inhomogeneous media, transit time calculation in propagating through parabolic index medium.
2	Refraction and Reflection by Spherical Surfaces	Basic concepts of Refraction at a Single Spherical Surface, Reflection by a Single Spherical Surface, The Thin Lens, The Principle Foci and Focal Lengths of a Lens, The Newton Formula, Lateral Magnification	T.B.: 3.1-3.7	To describe the image formation at spherical surface and to learn the formation of paraxial image by thin lens.
3	Matrix methods in paraxial optics.	Matrices for translation, refraction and reflection, unit planes, nodal points, system of two thin lenses.	T.B.: 4.1 – 4.5	Using the matrix method one calculate the effect of translation and refraction on paraxial rays. One can obtain the position of nodal points in an optical system and analyze the combination of lenses.



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4	Aberrations	Chromatic aberration, monochromatic aberrations, distortion	T.B.: 5.1-5.4	To understand that chromatic aberration and monochromatic aberrations.
5	Origin of refractive index.	Origin of refractive index; Dispersion: normal and anomalous dispersion. Rayleigh Scattering	T.B.: 6.5-6.6	To understand the origin of refractive index and describe the phenomenon of Rayleigh scattering
6	Group velocity and pulse dispersion	Group velocity of a wave packet; propagation in a non-dispersive and dispersive medium.	T.B.: 11.1-11.3	To explain the concept of group velocity, illustrate the group velocity of a wave packet using Gaussian pulse. To understand propagation in a dispersive and a non-dispersive medium.
7	Two beam interference by division of wavefront.	Interference of light waves, intensity distribution, Fresnel biprism. Displacement of fringes.	T.B.: 13.4-13.12	To understand the interference pattern, the intensity distribution are the foundations to the applications e.g. Fresnel biprism. Displacement of fringes, phase change on reflection.
8	Interference by division of amplitude	Interference by a plane parallel film, reflecting non- films, colors of thin films, Newton's rings	T.B.: 14.1-14.4, 14.10-14.11	To understand thin film interference; to explain formation of Newton's rings, working of Michelson Interferometer
9	Coherence	Temporal coherence and line width, spatial coherence.	T.B.: 17.1-17.4	To define and hence estimate coherence time and coherence length of a given source, estimate the monochromaticity or the spectral purity of source, concept of spatial coherence.
10	Fraunhofer Diffraction I	Fraunhofer diffraction: Single slit, double-slit, N- slit diffraction pattern, diffraction grating	T.B.: 17.1-17.3, 17.6 -17.8	Fraunhofer diffraction by single slit and double slit, a student will be able to apply the concept of N slit diffraction to understand the working principle of grating and dispersion power of grating, its applications.
11	Fresnel Diffraction	Fresnel diffraction: Fresnel half-period zones, the zone - plate	T.B.: 19.1-19.3	To analyze Fresnel half-period zones and its applications.
12	Polarization and double refraction.	Malus and Brewster's laws, double refraction, optical activity, linearly and circularly polarized light, Jones calculus.	T.B.: 21.1-21.7, 21.14-21.15	To learn to produce various forms of polarized light waves; how to control polarization of EM waves at interfaces with various angles of incidence, concept of Brewster angle and its importance in various applications, Jones calculus, Faraday rotation.
13	Laser: An Introduction	Spontaneous emission, Stimulated emission, Einstein coefficients, threshold condition, some laser systems.	T.B. 27.1-27.10	How stimulated emission is responsible for lasing action? components of laser and laser action. Explain fiber, Ruby and HE-NE lasers. Raman applications of laser.

6. Evaluation Scheme:

E.C.No.	Evaluation Scheme	Duration	Weightage (%)	Date, Time & Venue
1.	Mid Semester Exam (Open Book)	90 minutes.	30	05/10/2026, AN 1
2.	Tutorial Tests(Closed Book and Announced)	30 minutes.	20	To be announced in the class.
4.	Class discussions and Assignments linked to attendance		10	
3.	Comprehensive Exam (Closed Book)	3Hrs	40	03/12/2026, FN

7. Chamber Consultation Hour: Thursday 4-5 PM.

8. Notices: All important announcements will be made in the lecture classes and also put up in the Google classroom of the course.



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9. **Make-up Policy:** Make up will be given strictly to **genuine cases only** i.e. (i) Sickness leading to hospitalization, (ii) out of station with prior intimation & permission. It will be given for the mid-semester and comprehensive examination only.
10. **Attendance Policy:** Each student is expected to attend the classes and write all the components. Also, it is mandatory for the students to submit the assignments.

Niladri Sarkar
IC: PHY F213